

RETAIL INVENTORY & DEMAND FORECASTING ANALYSIS

Balancing Forecast Accuracy, Inventory Constraints, and Replenishment Strategy

A Statistical Analysis of Retail Demand Using R

By: Brianah Barnes



Business Question

How well does the retail system align forecasts, inventory, and replenishment with actual sales?

Forecasting

Does forecasted demand accurately reflect actual sales?

Inventory

Does limited inventory cap actual sales?

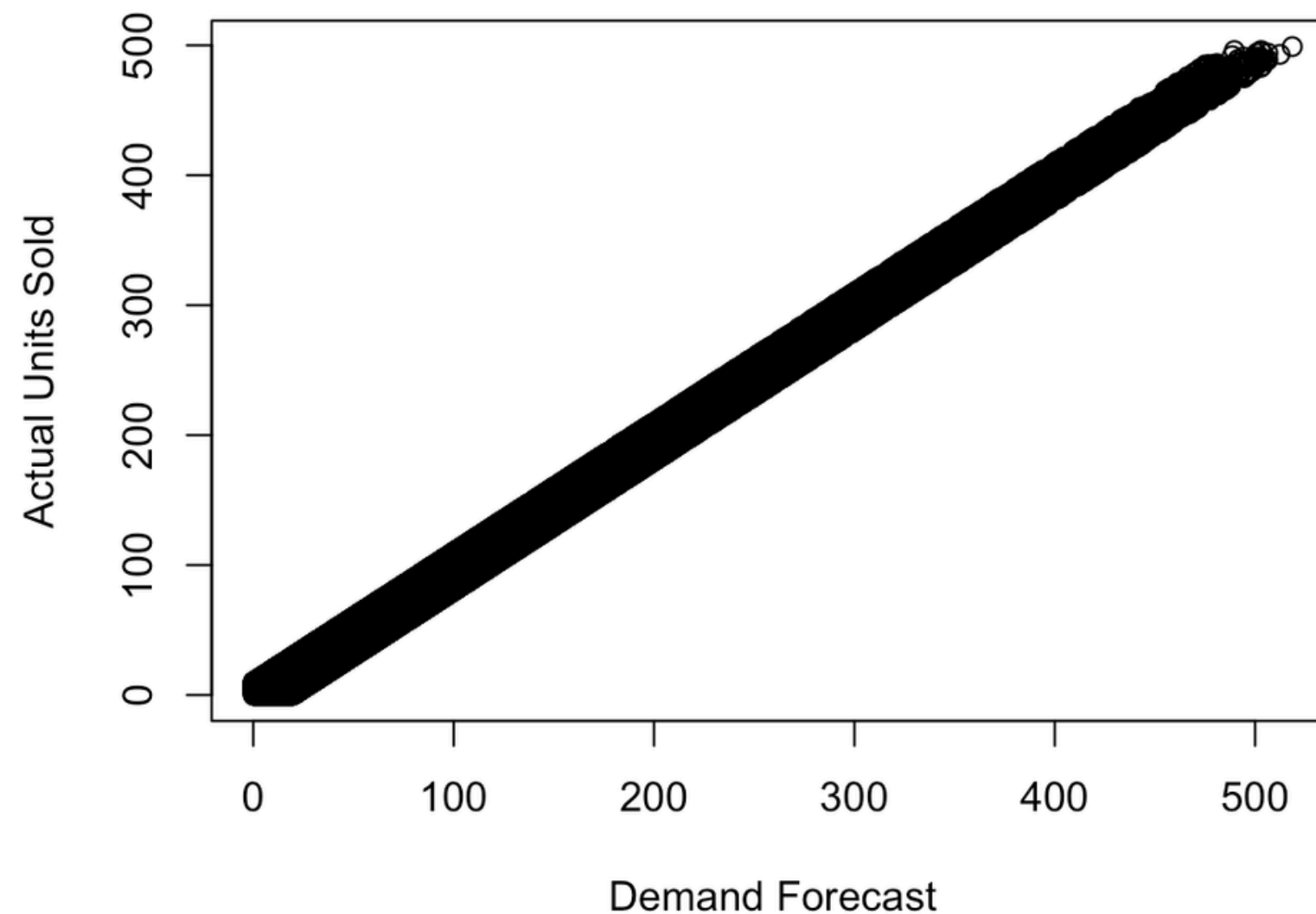
Replenishment

Are order quantities aligned with observed demand?

72,427 observations • 5 product categories • Statistical analysis using R

Forecast Performance

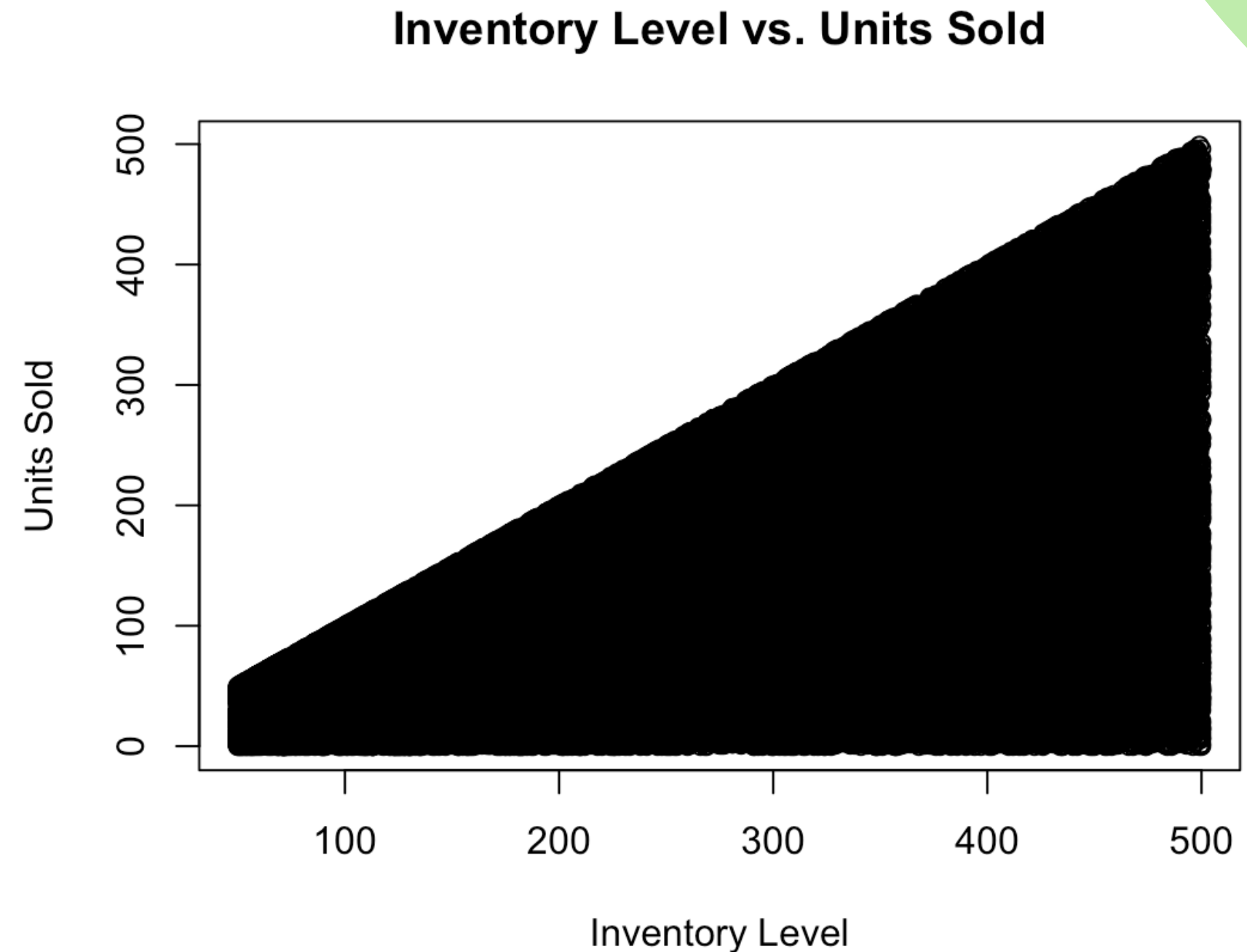
Demand Forecast vs Actual Sales



- **Correlation (r):** 0.997
- **Mean Forecast:** 142.84 units
- **Mean Actual Sales:** 137.71 units
- **Key takeaway:** Forecasts closely track actual sales, but overestimate demand by 5.14 units on average, leading to potential excess holding costs.

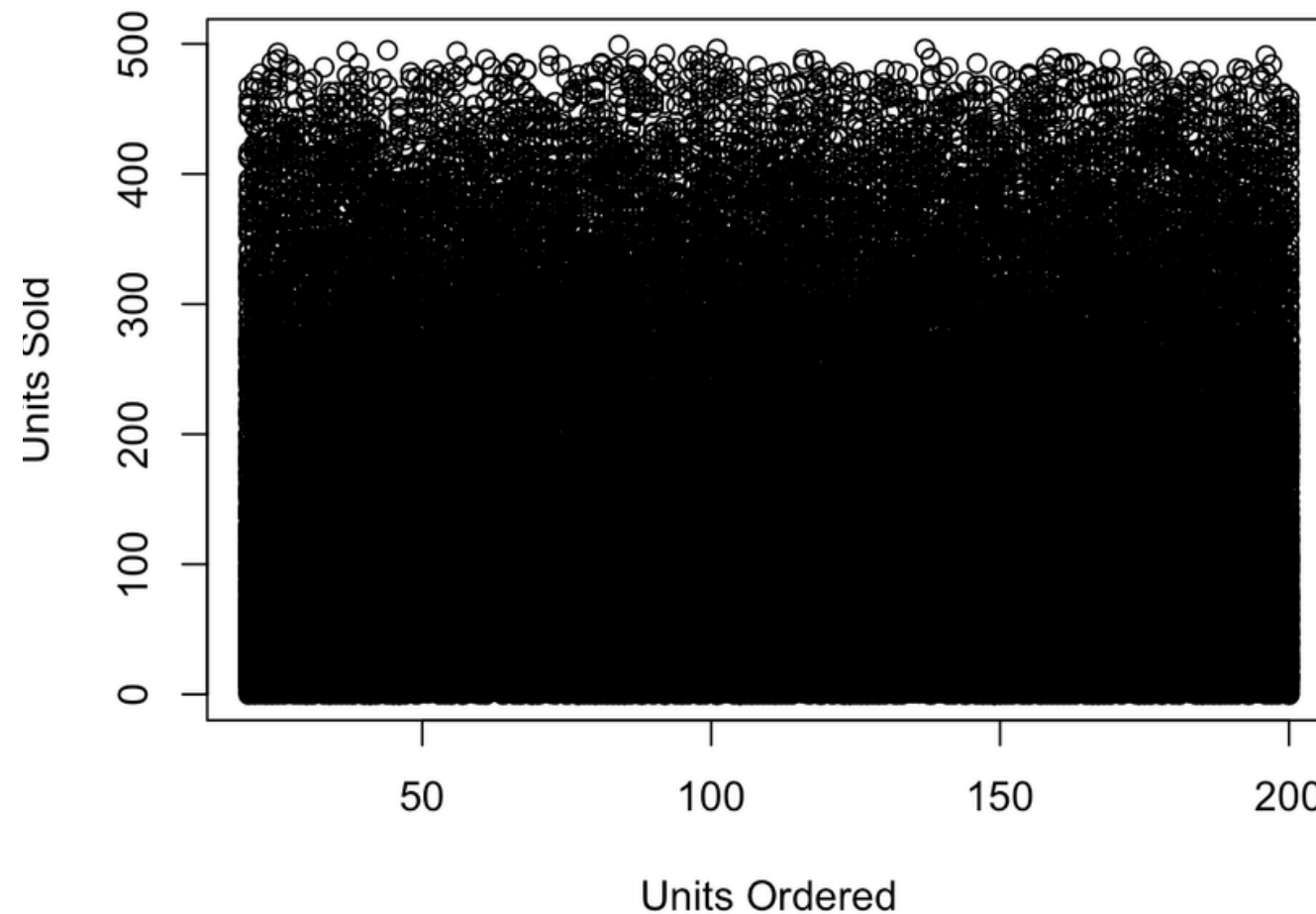
Inventory Constraint

- **Correlation (r):** 0.591
- **3.57%** of observations had forecast demand above available inventory.
- **Key takeaway:** Inventory can cap actual sales when demand exceeds available stock, as seen by the upper sales boundary at inventory levels.



Replenishment Alignment

Units Ordered vs. Actual Sales



- **Correlation (r):** -0.001
- **Key takeaway:** Order quantities show essentially no linear relationship with actual sales or forecast demand.

Key Strategic Question: Are replenishment quantities aligned with demand?

Key Findings & Recommendations



Forecasting

Forecasts are highly correlated with actual sales, but show a consistent upward bias, resulting in systematic overestimation of demand.



Inventory

Inventory can constrain sales when forecast demand exceeds available stock, limiting potential sales during peak periods.



Replenishment

Ordering quantities show little linear relationship with observed demand, suggesting that replenishment decisions may not be strongly responsive to demand levels.

Recommended Next Steps

- Use demand-sensing signals to make faster forecast adjustments
- Compare forecasted demand with inventory levels to reduce stockout risk
- Evaluate replenishment rules using both demand and inventory signals
- Incorporate external factors such as market trends, weather, and local events

From Exploration to Statistical Modeling

The exploratory analysis identified strong relationships between forecasts, inventory, and sales. The next step is to test whether inventory provides additional explanatory power beyond demand forecasts.

Units Sold ~ Demand Forecast

Model 1 — Baseline **$R^2 = 0.9937$**

Evaluates the baseline explanatory power of demand forecasts on actual point-of-sale volumes.

KEY RESULT

Adding inventory significantly improves model fit ($p < .001$), but the practical gain is minimal. R^2 remains 0.9937, showing that demand forecasts explain nearly all variation in sales.

Units Sold ~ Demand Forecast + Inventory Level

Model 2 — Extended **$R^2 = 0.9937$**

Tests whether inventory explains additional variation in sales, or if forecasts already capture it

THANK YOU!

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